Point Pattern and Polygon Analysis

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1.0 Introduction

This report presents the work for Exercise 2 of the course Intro to Spatial Data Analysis 2026. The exercise uses desktop GIS software to study point pattern and polygon data with spatial statistics. It has three problems. Problem 1 looks at where building fires happened in Helsinki from 2008 to 2010. Problem 2 looks at how population density is clustered in the Helsinki Capital Region, which covers Helsinki, Espoo, Vantaa and Kauniainen. Problem 3 gives a time estimate and feedback.

The fire data came as a text file in the old Finnish coordinate system KKJ3/YKJ (EPSG 2393), so it was first changed into a point layer. Two methods were then used on the points. Ripley's K function tests if the points are clustered, and Kernel Density draws a heat map of where the fires are concentrated. For Problem 2, the Finnish population grid with 1 km cells and the Paavo postal code areas were used. Population density was studied with the global Moran's I test and the Anselin Local Moran's I test.

The main results are as follows. The building fires are clearly clustered at all distances up to about 8,300 m, and the clustering is strongest in central Helsinki. The kernel density maps show that the search radius changes the heat map a lot, while the cell size has little effect. Both population datasets show strong positive spatial autocorrelation, with a Moran's Index of 0.572422 for the postal areas and 0.500874 for the grid cells. The sections below explain each step, show the main figures and answer the questions of the exercise.

Problem 1

2.0 Task 1.1: Fire Points, Ripley's K and Kernel Density

2.1 Point layer

The building fires data for 2008 to 2010 was downloaded from the course GitHub as a CSV file. The XY Table To Point tool turned the table into points, and the coordinate system was set to KKJ Finland Uniform Coordinate System (EPSG 2393). A basemap was added to check the result. The points fall in the right place over Helsinki and the nearby cities, and most of them are in the city centre (Fig 1).

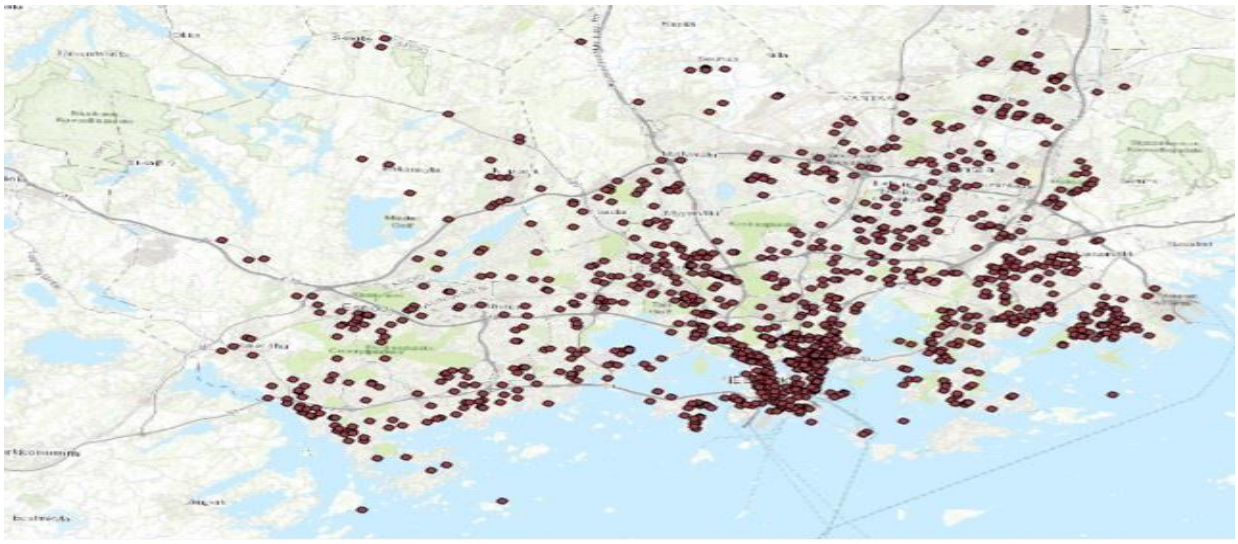


Fig 1: Helsinki building fires from 2008 to 2010 (KKJ3/YKJ, EPSG 2393) shown over a basemap

2.2 Ripley's K function

Ripley's K function tests if the fire points are clustered, random or evenly spread at many distances. The Multi-Distance Spatial Cluster Analysis tool was run two times on the fire points, because the exercise asks to check if the number of distance bands changes the result. The first run used 10 distance bands and the second run used 30. The weight field was left empty, since every point is one fire. Both runs used a confidence envelope from 99 permutations, so the result shows if the clustering is significant and not only visible. Fig 2 and Fig 3 show the two K function charts.

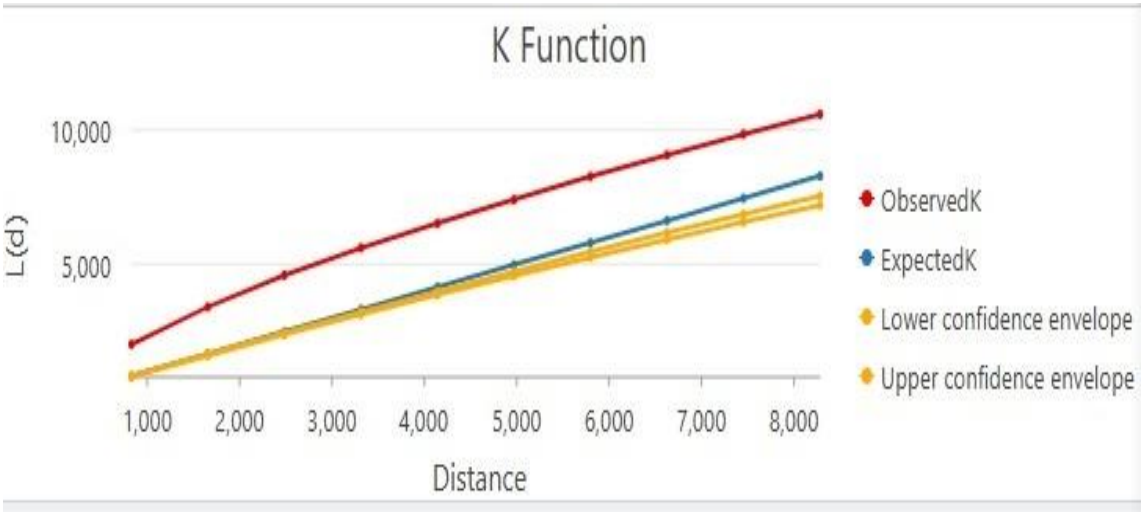


Fig 2: Ripley's K function for the building fires with 10 distance bands and a confidence envelope from 99 permutations

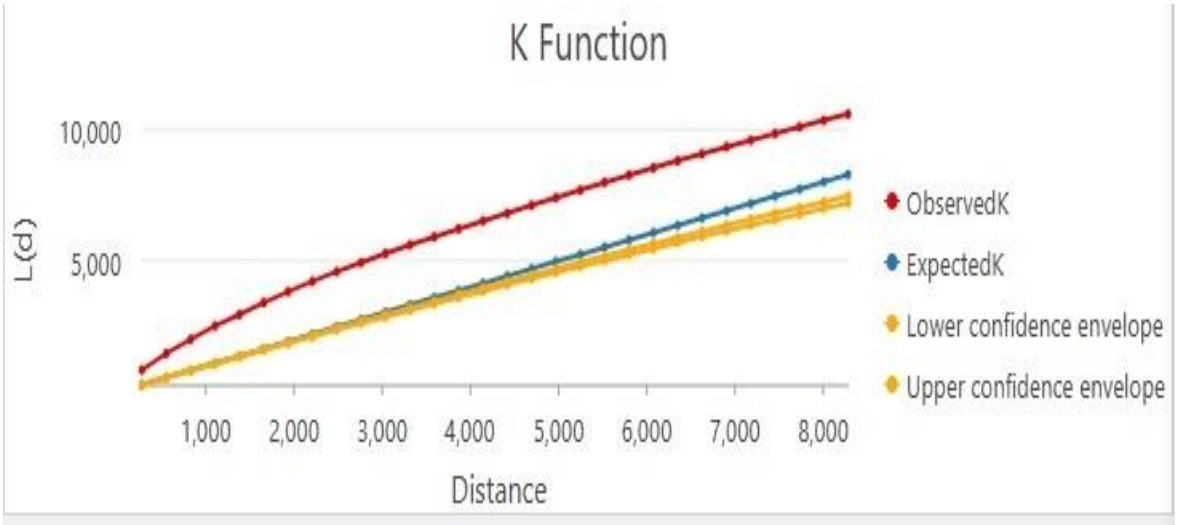


Fig 3: Ripley's K function for the building fires with 30 distance bands and a confidence envelope from 99 permutations

Both charts give the same result. The observed K line is above the expected line and above the confidence envelope at every distance, up to about 8,300 m, so the fires are clustered at all the distances tested. The 30 band chart has steps of about 280 m instead of 830 m, so the curve is smoother and it shows that the clustering is already strong at the shortest distances. The number of distance bands therefore changes the detail of the curve but not the conclusion. The answers to Q1.1 and Q1.2 explain these results in more detail.

2.3 Kernel density with two cell sizes

Kernel Density was run with the Planar method and the Densities setting for the cell values. The population field was set to none, and the area unit was square meters. Every map uses five Natural Breaks classes and the same colour scheme, so the maps can be compared by eye. Fig 4 and Fig 5 use the same search radius of 2000 m with cell sizes of 50 m and 100 m.

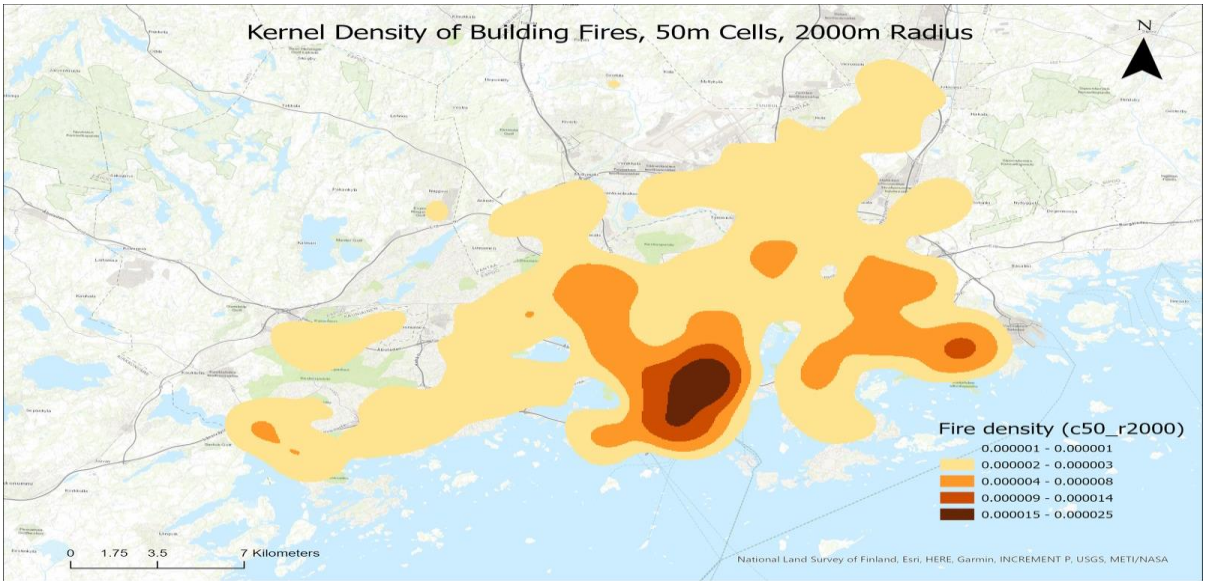


Fig 4: Kernel density of building fires with 50 m cells and a 2000 m search radius (highest class 0.000015 to 0.000025 fires per square meter)

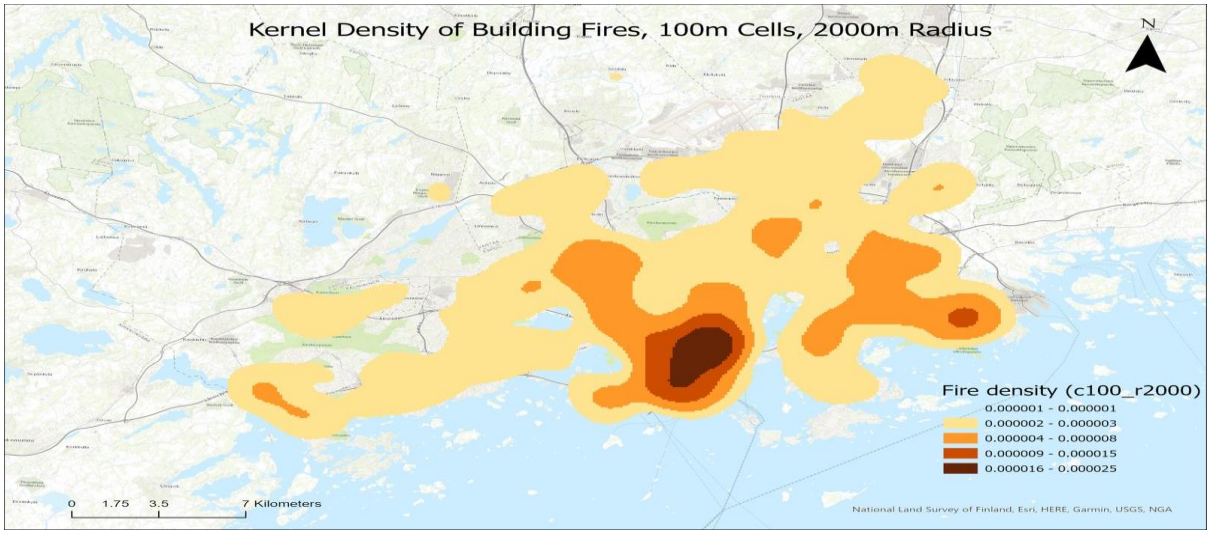


Fig 5: Kernel density of building fires with 100 m cells and a 2000 m search radius (highest class 0.000016 to 0.000025 fires per square meter)

2.4 Kernel Density with Three Search Radii

Three search radii of 500 m, 2000 m and 5000 m were tested with the same cell size of 50 m. The map with the 2000 m radius is Fig 4, so only the 500 m and 5000 m maps are shown below (Fig 6 and Fig 7). Table 1 lists the highest-class value of each map.

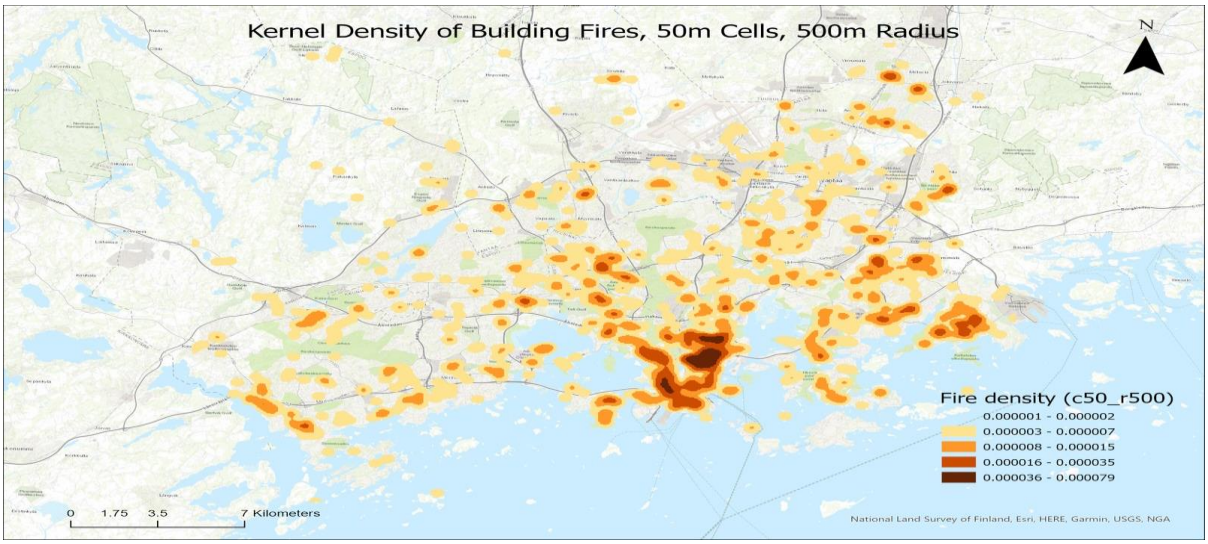


Fig 6: Kernel density of building fires with 50 m cells and a 500 m search radius (highest class 0.000036 to 0.000079 fires per square meter)

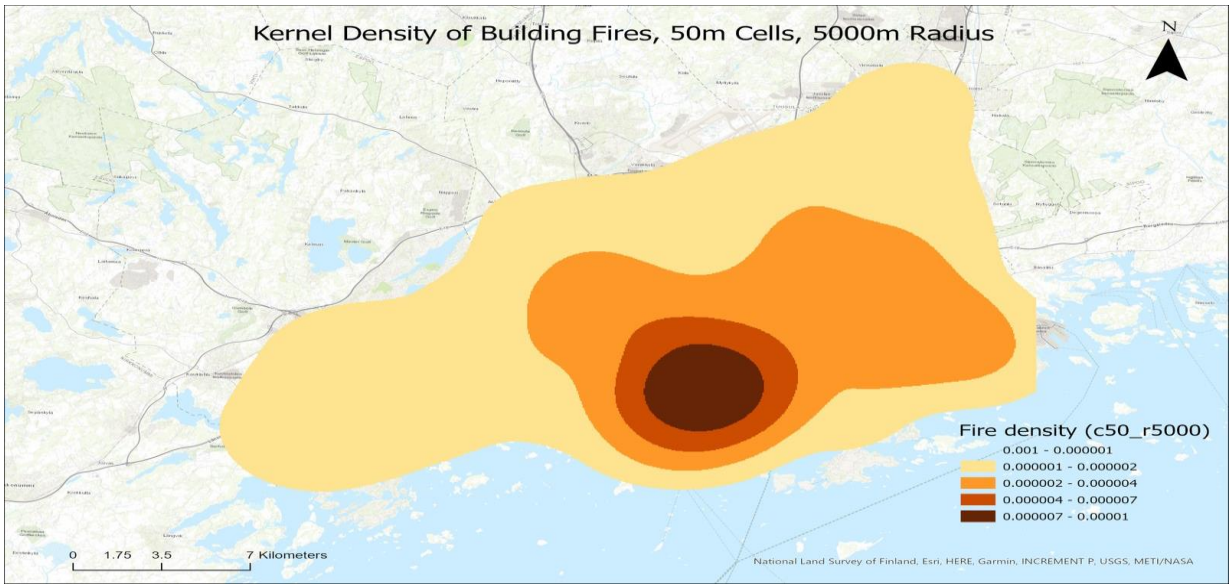


Fig 7: Kernel density of building fires with 50 m cells and a 5000 m search radius (highest class 0.000007 to 0.00001 fires per square meter)

Table 1: Highest class value of each kernel density map, in fires per square meter

Raster	Cell size	Search radius	Highest value
c50_r500	50 m	500 m	0.000079
c50_r2000	50 m	2000 m	0.000025
c100_r2000	100 m	2000 m	0.000025
c50_r5000	50 m	5000 m	0.00001

3.0 Task 1.2: Questions on Ripley's K and Kernel Density

Q1.1: The observed K line is above the expected K line at every distance, and it is also above the confidence envelope. This is true for both the 10 band chart and the 30 band chart, up to about 8,300 m. So the building fires are clustered and not random. In the 10 band chart, the observed value at about 830 m is about 2,000, while the expected value is about 830. At about 8,300 m the observed value is about 10,600 and the expected value is about 8,300. The observed value is about 2.4 times the expected

value at the shortest distance and about 1.3 times at the longest distance, so the clustering is strongest at short distances. This fits the point map, where most fires are in central Helsinki and in the built areas around it. Part of this clustering is natural, because fires only happen where buildings are, and the study area also has sea and empty land.

Q1.2: When ExpectedK is smaller than ObservedK, each point has more neighbours within that distance than a random pattern would give, so the points are clustered. When ExpectedK is larger than ObservedK, each point has fewer neighbours than expected, so the points are more evenly spread than random. This is called dispersion. When the two lines are close, or the observed line is inside the confidence envelope, the pattern is close to random.

Q1.3: The search radius controls how smooth the map is. A small radius of 500 m gives many small and sharp hotspots in different neighbourhoods (Fig 6). A radius of 2000 m joins these spots into wider zones, with one main centre and smaller zones in the west and the east (Fig 4). A radius of 5000 m gives one smooth area around the city centre, and the local detail is gone (Fig 7). The output cell size only changes the size of the squares in the grid. The maps with 50 m and 100 m cells look almost the same, and the 100 m map has slightly more blocky edges (Fig 4 and Fig 5).

Q1.4: The values use the same unit, but they are not fully comparable. A bigger radius spreads the same fires over a bigger area, so the peak value falls from 0.000079 at 500 m to 0.000025 at 2000 m and 0.00001 at 5000 m (Table 1). Each map also has its own class limits, because Natural Breaks was used, so the same colour means a different value on each map. Dark brown on the 5000 m map shows far fewer fires per square meter than dark brown on the 500 m map. The maps are comparable in pattern, because the city centre is the highest area on all three, but they are not comparable in values or colours. A fair comparison would need the same fixed class limits for all maps.

Problem 2

4.0 Task 2.1: Spatial Autocorrelation and Cluster Analysis

4.1 Data preparation

The municipality layer was used to select Helsinki, Espoo, Vantaa and Kauniainen. The grid cells and postal areas that touch these four cities were then selected, which gave 805 grid cells and 185 postal areas. The 185 postal areas were exported to a new layer called paavo_hki, and a new Double field called popdens was added. It was calculated as he_vakiy divided by pinta_ala, which gives people per square meter. For the grid, the population field vaesto was used directly, because it holds the number of people in each 1 km cell and so no new field was needed.

4.2 Global Moran's I

The Spatial Autocorrelation (Global Moran's I) tool was run once for each dataset, with Contiguity edges corners and Row standardization. Table 2 shows the results. The Moran's Index is 0.572422 for the postal areas, with a z-score of 12.017838. It is 0.500874 for the grid cells, with a z-score of 25.675818. Both p-values are 0, so both results are significant. Each run gave a warning that some features have no neighbors, which affected six postal areas and three grid cells.

Table 2: Global Moran's I results for the postal areas and the population grid

Measure	Postal areas	Population grid
Number of features	185	805
Moran's Index	0.572422	0.500874
z-score	12.017838	25.675818
p-value	0.000000	0 (as reported)
Moran's scatterplot R^2	0.54	0.46
Features with no neighbors	6	3

4.3 Cluster and Outlier Analysis

The Cluster and Outlier Analysis (Anselin Local Moran's I) tool was run with the same Contiguity edges corners and Row standardization settings and with 499 permutations. Fig 8 and Fig 10 show the cluster maps, and Fig 9 and Fig 11 show the Moran's scatterplots for the postal areas and the grid cells.

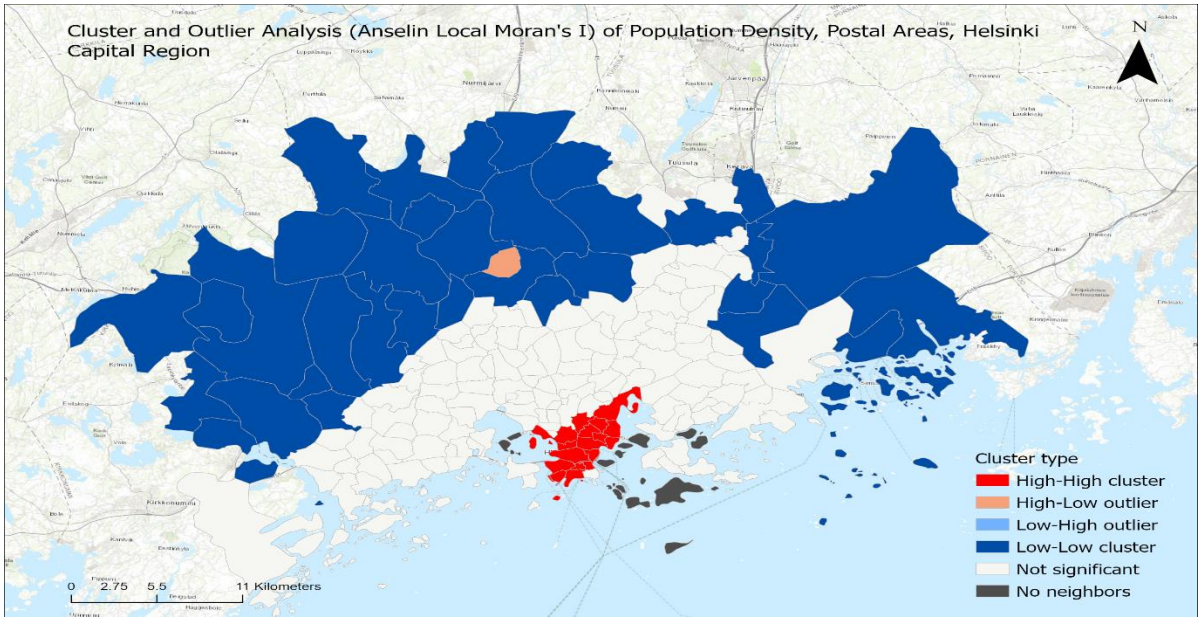


Fig 8: Cluster and Outlier Analysis (Anselin Local Moran's I) of population density for the 185 postal areas of the Helsinki Capital Region (the dark grey areas have no neighbors)

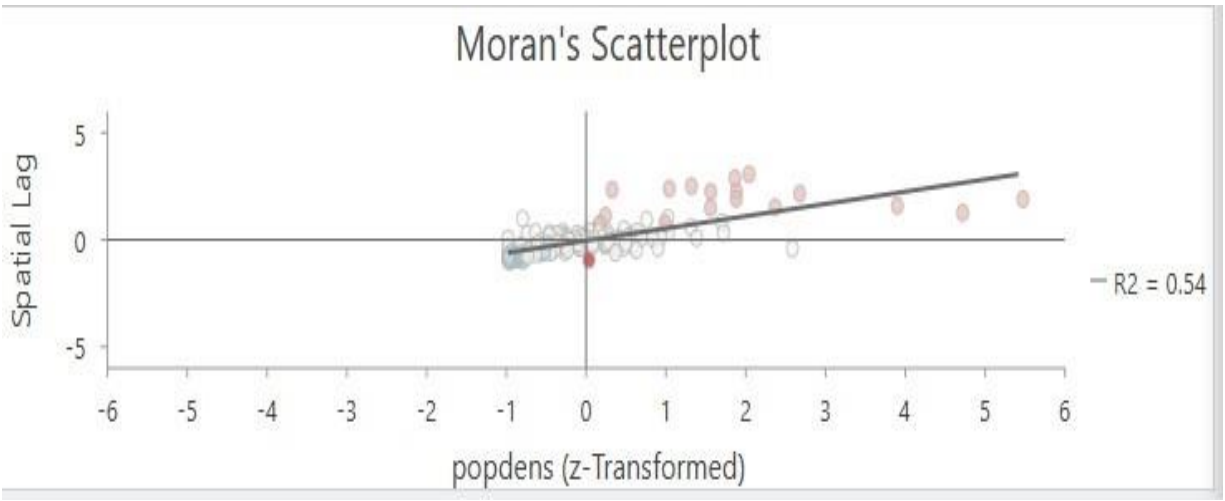


Fig 9: Moran's scatterplot of population density for the postal areas (R^2 is 0.54)

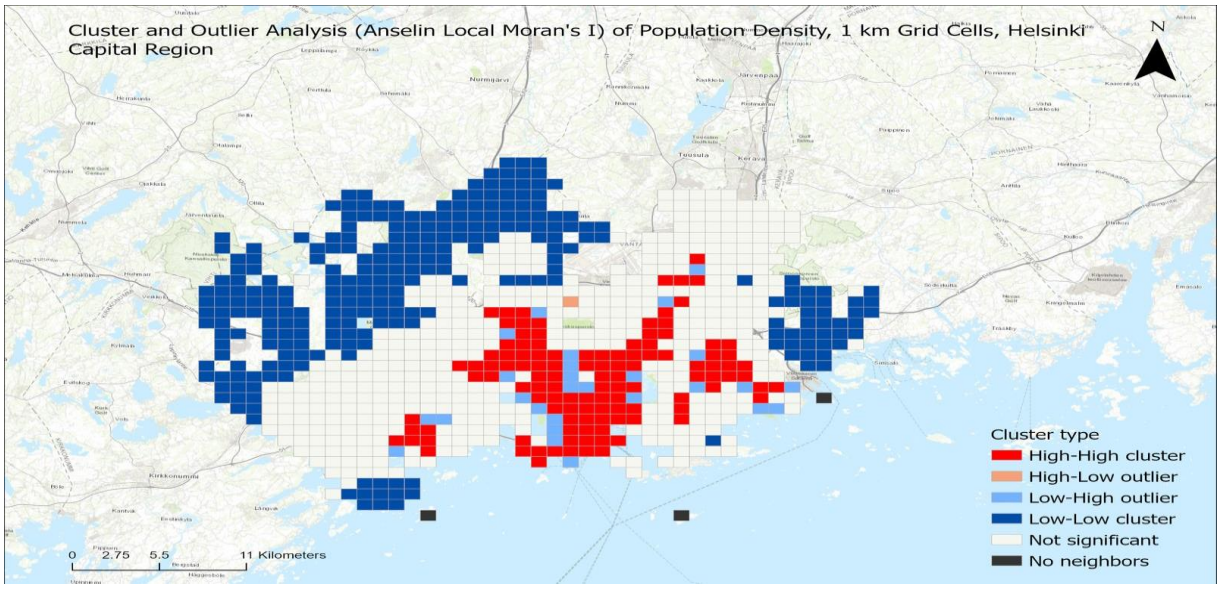


Fig 10: Cluster and Outlier Analysis (Anselin Local Moran's I) of population for the 805 grid cells of 1 km by 1 km in the Helsinki Capital Region (the dark grey cells have no neighbors)

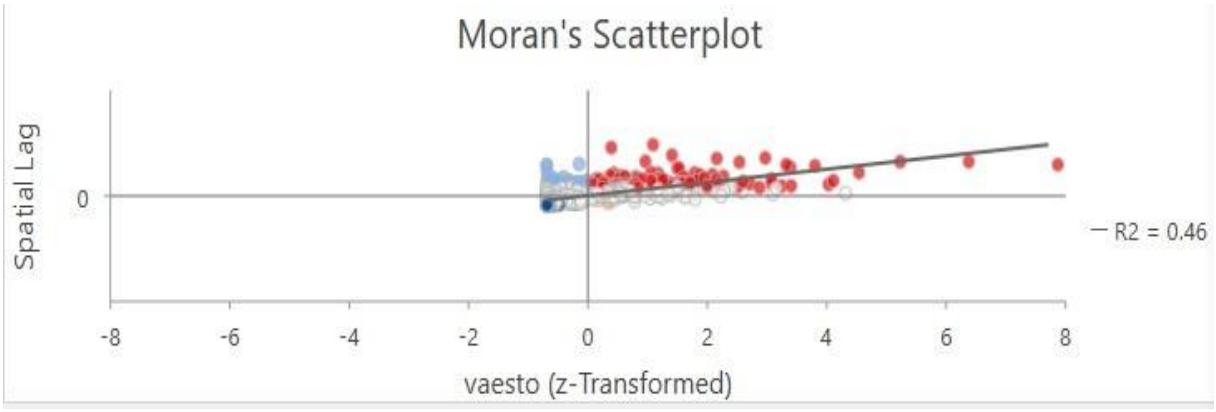


Fig 11: Moran's scatterplot of population for the grid cells (R^2 is 0.46)

4.4 High-Low outlier cell in the population grid

The grid has one High-Low outlier cell, and it has a population (vaesto) of 2353 people. The cell is in the Ylästö postal area (postcode 01690) in Vantaa (municipality code 092). Fig 12 shows the cell and its surroundings, and Fig 13 shows a closer view inside the cell.



Fig 12: The High-Low outlier cell (vaesto is 2353) and its surroundings on an imagery basemap, shown as the square with the orange tint in Ylästö, Vantaa



Fig 13: A closer view inside the High-Low outlier cell, showing low-rise housing, a few larger buildings and a sports field

5.0 Problem 2, Task 2.2: Questions on Moran's I

Q2.1: The Moran's Index is 0.572422 for the postal areas and 0.500874 for the grid cells (Table 2). Both values are positive and far above the expected value, which is close to zero. The z-scores are 12.017838 and 25.675818, and both p-values are 0, so the results are very significant. This means that both datasets are spatially clustered. Areas with high population density are next to other areas with high density, and areas with low density are next to other areas with low density. The scatterplots show the same thing, because most points are in the high-high and low-low parts of the plot and the trend line goes up (Fig 9 and Fig 11).

Q2.2: LMiZScore_RS is the local z-score of each postal area or grid cell. A high positive value means the area is similar to its neighbours, so it is part of a cluster. A high negative value means the area is different from its neighbours, so it is an outlier. LMiPValue_RS is the p-value of that local result, and it tells how likely it is that the pattern happened by chance. A small value, such as below 0.05, means the result is significant, and the area then gets a cluster or outlier type in the COType_RS field. The RS ending means that row standardization was used.

Q2.3: The High-Low outlier cell is in the Ylästö postal area in Vantaa, and it has 2353 residents. The cell is a compact area of low-rise houses, a few larger buildings and a sports field (Fig 12 and Fig 13). Around it there is mostly forest, a lake and farmland, where very few people live. So a built area with many people sits inside a green area with few people. The test marks the cell as High-Low because its value is high and the average value of its neighbours is low.

Q2.4: Both maps show the same overall pattern (Fig 8 and Fig 10). There is a High-High cluster in the centre of Helsinki and in the dense areas around it. There are Low-Low clusters in the outer parts of the region in the north, the west and the east, where there are forests, fields and few houses. Most areas in between are not significant, and there are only a few outliers, such as one High-Low outlier in each dataset. This result is realistic, because a city is usually dense in the centre and becomes less dense towards the edge. It also matches what is known about the capital region.

There are some limits to this result. The postal areas have very different sizes, so large outer areas can look alike only because they are large. Six postal areas and three grid cells have no neighbors, so they could not be tested. Cells at the edge of the region have fewer neighbours, because the area outside the four cities was not included. Some grid cells show the value 1 with matching columns marked as missing, which may mean that very small counts are hidden.

Q2.5: Both datasets show clear spatial clustering, but there are some differences (Table 2). The Moran's Index is a little higher for the postal areas (0.572422) than for the grid (0.500874), and the scatterplot R^2 is also higher (0.54 and 0.46). The z-score is much higher for the grid (25.675818 against 12.017838). The reason is that the grid has 805 cells and the postal data has only 185 areas, so the grid test has more evidence.

The maps differ too. Postal areas are small in the centre and large in the outer parts, so neighbours with similar density are grouped together, and this raises the index. The grid cells have equal size, so the grid shows more detail. It marks High-High cells in other parts of Helsinki and north of the centre, and it has more Low-High cells near the central cluster. The main conclusion is the same for both datasets, but the size and shape of the areas change the result a little.

6.0 Problem 3: Time Estimate

The exercise took about 6 hours in total. This time covered reading the instructions, preparing the data, running the point pattern and Moran's I tools, testing different kernel density settings, and building the maps. Most of the time went on the kernel density maps and the layouts, because several versions had to be made and compared.

7.0 Feedback

The exercise gave a clear view of how spatial statistics work at three levels, which are points, postal areas and grid cells. Moving from Ripley's K function and kernel density to Moran's I showed how each method answers a different question about the same city. The maps and charts made the ideas of clustering and outliers easy to follow, and testing different search radii showed clearly how one setting can change a map.

The exercise was well organised. The three problems built on each other, and the instructions were clear about the tools and settings to use. Working with real data from the Helsinki Capital Region made the results meaningful, because the patterns could be checked against places that are known.